

Supplementary Materials for “Neural representations of reward-related memories shift across development” (Cohen*, Benear*, et al., *Journal of Neuroscience*)

FIGURES

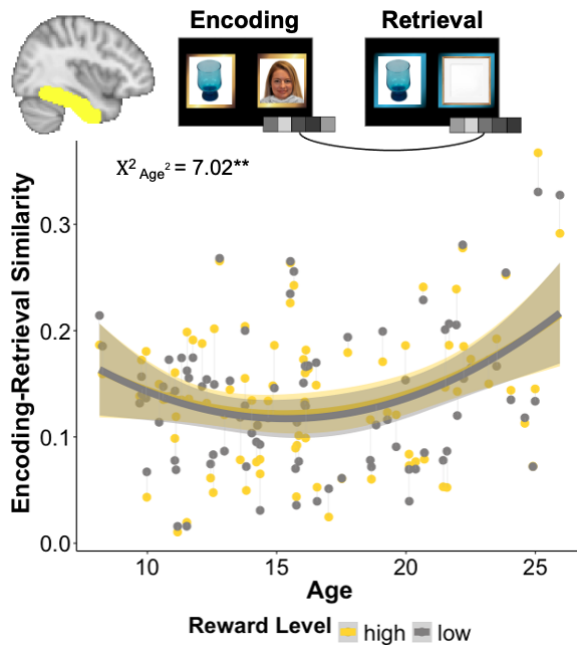


Figure S1. Encoding-retrieval similarity (ERS) in ventral temporal cortex (VTC) by reward level and age. VTC ERS does not vary by reward level, but changes non-linearly with age, indicated by a significant age² term ($\chi^2 = 7.02$, $p = .008$). Specifically, VTC ERS generally increases with age, with the quadratic parameter capturing an acceleration of change in early adulthood. **** $p < .01$**

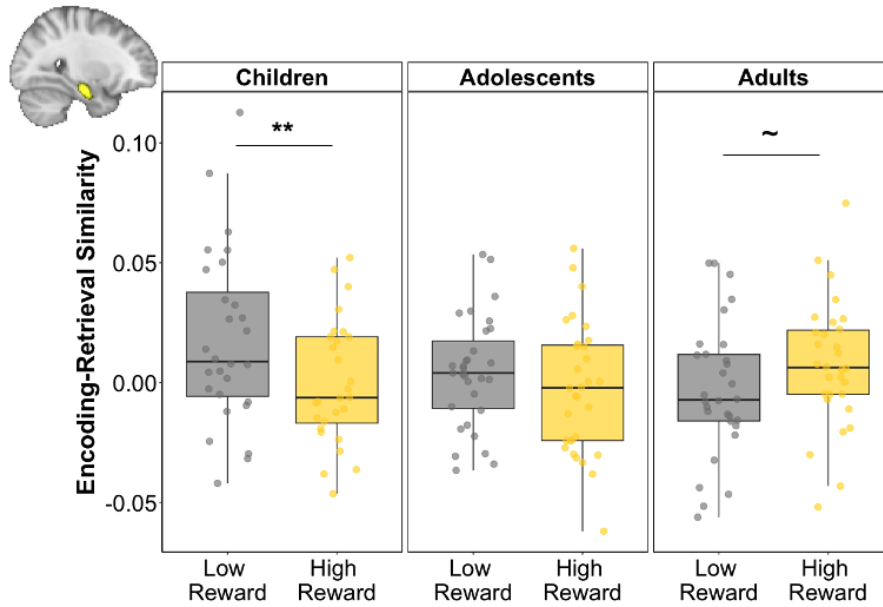


Figure S2. Encoding-retrieval similarity (ERS) in anterior hippocampus (aHC) by reward level and age group. When re-running our mixed-effects model predicting aHC ERS by age group (children, age 8-12.99; adolescents, age 13-17.99; and adults, age 18-26) rather than continuous age, there was still a significant interaction between reward level and age group (Adults vs. Adolescents: $\beta = .017$, $p = .046$; Adults vs. Children: $\beta = .029$, $p = .002$). When examining the estimated marginal means separately for each group, we found that low-reward ERS is greater than high-reward ERS for children ($\beta = -.018$, $p = .006$), high-reward ERS shows a trend toward being greater than low-reward ERS for adults ($\beta = .011$, $p = .092$), and low-reward ERS is not significantly different from high-reward ERS for adolescents ($\beta = -.007$, $p = .27$). The model was run at the trial level with a random intercept for each participant, but participant-level averages are plotted. ** $p < .01$, ~ $p < .10$

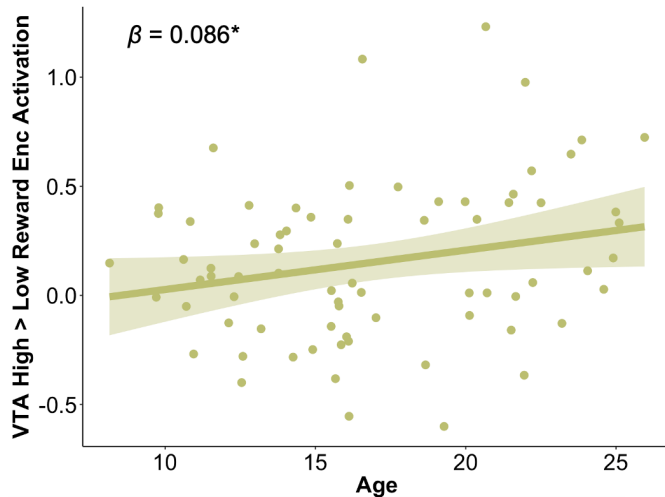


Figure S3. The relationship between age and univariate activation in ventral tegmental area (VTA). There is a significant relationship between age and VTA high versus low univariate activation ($\beta = .086$, $p = .043$), demonstrating that value-based signaling generally increases with age across childhood and adolescence into young adulthood.

TABLES

Table S1. Brain-behavior analyses.

Model and corresponding main text figure	Model code
Trial-wise generalized linear mixed-effects analysis, implemented using the <i>glmer</i> function in R (Figure 3)	<i>memory ~ VTC ERS scaled + reward level + age scaled + high reward stim category + (reward level subject ID)</i>
Multiple regression model 1, implemented using the <i>lm</i> function in R (Figure 4A & B)	<i>high – low memory ~ high – low aHC ERS + high – low VTC encoding similarity + age scaled + high reward stim category</i>
Multiple regression model 2, implemented using the <i>lm</i> function in R (Figure 5)	<i>high – low aHC ERS ~ high – low VTA encoding activation + age scaled + high reward stim category</i>

For table S2, the dependent variable for each model comes from an outcome measure in the columns paired with a brain region in the row (e.g. encoding similarity in VTC), with the exception of behavioral memory, for which there is no associated region. The predictors in all models are age, reward level (high vs. low), retrieval condition, and high-reward source image category.

Table S2. Age and reward effects on behavior and neural pattern similarity measures.

	Effect of age on:			
	Encoding Similarity	Retrieval Similarity	Encoding-Retrieval Similarity (ERS)	Specific source memory
Ventral Temporal Cortex (VTC)	$\chi^2(1, N = 73) = 4.38, p = 0.036$	NS	<i>Effect for age²: $\chi^2(1, N = 89) = 7.02, p = 0.008$</i>	NS
Anterior Hippocampus (AHC)	NS	NS	<i>Interaction with reward level: $\chi^2(1, N = 89) = 9.41, p = 0.002$</i>	
	Effect of reward level (high versus low) on:			
	Encoding Similarity	Retrieval Similarity	Encoding-Retrieval Similarity (ERS)	Specific source memory
Ventral Temporal Cortex (VTC)	$\chi^2(1, N = 73) = 13.85, p = 0.002$	NS	NS	$\chi^2(1, N = 89) = 17.89, p < 0.001$
Anterior Hippocampus (AHC)	NS	NS	<i>Interaction with age: $\chi^2(1, N = 89) = 9.41, p = 0.002$</i>	